RESEARCH BRIEFING · FOR PRESENTATION

How the 60-DTE implied-volatility relative-value model works

A walk-through of the data, smile models, residual, z-gate, TRADE / NO TRADE / CANNOT DECIDE rule, and how fair those calls really are.

Item	Value
Research / train / tune	2025-06-05 → 2026-09-08
Unused gap	2026-09-09 and 2026-09-10
Frozen holdout	2026-09-11 (signals only, no retuning)
Target tenor	Nearest single expiry to 60 calendar DTE
Frozen production gate	z = 3.0 on the 60-day residual
Dashboard	https://snp-60dte-iv-desk.vercel.app
Repo	github.com/Ayushkumarsingh09/snp-60dte-iv-relative-value

One-line thesis: cheap is not low IV, and cheap is not "below SVI". A trade is only a residual that is historically extreme, agrees with its own sign, is not extrapolated, and still looks small versus costs. On a quiet day the honest answer is NO TRADE.

1. What this project is — and is not

This is a research engine and a static desk for **relative-value on a single ~60-day implied-volatility slice** of SNP / S&P-like option chains. Each trading day we take the last snapshot, pick the expiry nearest to 60 DTE, fit several smiles, measure how the market differs from a fair smile, standardise that difference against the structure's own history, and then decide **TRADE**, **NO TRADE**, or **CANNOT DECIDE**.

The objective is predictive relative value, not a pretty surface.

A smile that interpolates every listed strike with a tiny RMSE can still be useless for trading. SVI has the lowest in-sample IV RMSE (~0.47 vol points) but that is *not* why we use it. We care whether a residual today tells us anything about residual change over the next few days, and whether that change is large enough to survive a conservative cost model.

- **It is** a causal, walk-forward, holdout-protected study of 60-DTE IV residuals and structures.
- **It is** a dashboard that can score any RR, butterfly, slope, delta-point or listed strike.
- **It is not** a live execution system. There is no bid/ask, volume or open interest in the files.
- **It is not** a claim that 11 Sep 2026 had a trade. The frozen book said NO TRADE.
- **It is not** a calendar-consistent SSVI surface. One maturity cannot identify that.

The three sentences you can say in a meeting

- We compare market IV (or a structure built from it) to an ensemble fair value, not to a single parametric smile.
- We only call TRADE when that residual is historically extreme at a frozen z-gate of 3.0 and the signs agree.
- Residuals do mean-revert a little (5-day correlation about –0.44), but the typical size is 1–10 basis points of vol and does not survive modelled costs.

2. Research calendar — how we stop ourselves cheating

Every threshold, feature and model choice was locked on data through 8 September 2026. The holdout day is used only to *apply* the frozen rule. Later dates may score that decision; they must not reopen the gate.

Window	Dates	Allowed use
Research	2025-06-05 … 2026-09-08	Fit smiles, compute z, choose gate, train ML, backtest
Gap	2026-09-09, 2026-09-10	Present on disk; unused for train or tune
Holdout	2026-09-11	Generate signals with the frozen rule only
After test	2026-09-14 onward	Score the frozen NO TRADE / TRADE call

Figure-style reminder: 11 Sep 2026 must never change the gate.

Walk-forward inside the research window uses expanding training, ~20-day validation blocks, and a 5-day embargo so that a residual used as a feature is not paired with a label that leaked from the next fold. There are no random train/test splits.

3. What the data actually is

324 daily parquet files, weekdays only, 2025-06-05 to 2026-09-22. A typical file has ~12,000–13,000 rows, hourly stamps 09:00–16:00, ~43 expiries and ~80–90 strikes. We inspected the files first; we did not trust the brief when the files disagreed.

Field / fact	What we found and what we do
Column name	Files say **contract**, not contact.
Snapshot	Last timestamp of the day, usually 16:00. Early closes keep the last print.
rate	Identically 0. Using $F = S \cdot e^{rT}$ would give F = S, which is wrong.

Field / fact	What we found and what we do
Forward	Put-call parity near ATM: F = median(K + C – P). Discount factor = 1.
Underlying	Treated as spot. PCP carry is ~30–40 index points at ~57 DTE.
IV units	Decimal (0.16 = 16%). Vendor IV kept after a BS reconstruction audit.
put_delta	Unsigned to 2026-07-15, signed after. Engine always uses abs(put_delta).
atm	Integer strike offset (0 = closest listed strike), not a boolean.
DTE	Calendar days. Exact DTE = 60 on only 36/317 research+test days.
Missing	No bid, ask, volume, OI, dividends or event calendar.

Cleaning is logged, not silent: IV bounds 2%–150%, minimum 8 usable strikes, reconstruction audit on selected days. Across snapshots, 517,395 rows were seen, 504,690 kept, 12,705 removed. Call IV reconstruction MAE median is about 0.46 vol points on audit days — close enough in the centre, worse in the wings, which we filter.

4. Building one day’s 60-DTE surface

Step-by-step for every date t

- Load the parquet for date t and keep the last snapshot.
- Clean the full chain and record every drop.
- Select **one** expiry: exact 60 if it exists, otherwise the nearest expiry to 60. Never mix expiries across strikes. Mean |DTE–60| ≈ 2.85 days, max 8.
- Infer the PCP forward near ATM.
- Year-fraction T = DTE / 365. Log-moneyness k = ln(K / F).
- OTM IV rule: K < F uses put IV; K > F uses call IV; |k| < 0.01 blends them.
- Build total variance w = IV² T. That is what SVI / SSVI / the polynomial actually fit.
- Fit five smiles. Fair IV is the **median** of the five. Residual = market IV – ensemble fair IV.
- Interpolate a delta grid and build structures (RR, BF, slopes, points, verticals, listed strikes).
- Update rolling z-scores using only dates ≤ t.

On 11 Sep 2026 this produces: snapshot 16:00, expiry 2026-11-05 (OWSNP1EX26), actual DTE 56, spot 7661, PCP forward ≈ 7700, ATM IV 13.45%, 25-delta risk reversal –5.14 vol points.

5. The five smile models and the ensemble

Each model is a different opinion of “what a 60-DTE smile should look like”. A large residual where they agree is more interesting than a large residual where they fight.

Method	What it fits	Role
Quadratic polynomial	Total variance vs k: $w(k) \approx a + b\,k + c\,k^2$	Smooth global parabola; weak in the wings.
Smoothing spline	Flexible w(k) or IV(k), vega-weighted	Local shape; can overfit noise.
SVI (Gatheral)	$w(k) = a + b\{\rho(k-m) + \sqrt{((k-m)^2+\sigma^2)}\}$	Best in-sample RMSE (~0.00047). Still not the decision rule.
SSVI slice	θ, ρ, ϕ on a single maturity	Arbitrage-aware slice. Cannot be a full calendar SSVI here.
SABR, $\beta = 1$	α, ρ, ν on IV(K) at fixed T	Lognormal backbone; good near ATM, stylised wings.
Ensemble	Median of the five fair IVs	The official fair value. Dispersion = cross-model standard deviation.

Weights for the fits are vega by default (liquidity and hybrid weights exist). Previous-day SVI parameters warm-start the next research day, but the holdout day does not inherit a non-research initialisation. In-sample RMSE ranking is SVI < SABR < spline < SSVI < polynomial. We ignore that ranking when deciding trades.

Residual — the central object

```
residual  =  market_value  –  ensemble_fair_value
For a strike:   residual = market_IV – median(poly, spline, SVI, SSVI, SABR)
For 25D RR:     residual = (IV_25C – IV_25P)_mkt  –  (IV_25C – IV_25P)_fair
```

Sign convention: residual < 0 means the market is **cheap versus the ensemble** (below fair). Residual > 0 means **expensive versus the ensemble**. That is only Definition 1. It is not yet a trade.

6. Structures the model can talk about

A structure is any repeatable combination of smile points. Raw strike 7700 is not a time series — the forward moves — so listed-strike history is mapped to a log-moneyness bin of width 0.02.

Research-period book (pipeline)

- ATM IV level.
- Risk reversals RR_d = IV_dC – IV_dP at 10 / 15 / 20 / 25 / 30 delta.
- Butterflies BF_d = ½(IV_dP + IV_dC) – IV_ATM at the same deltas.
- Put / call skew vs ATM.
- Slopes ATM → wing (sometimes scaled by delta gap).
- Call and put verticals at pairs (25/15, 30/20, 20/10, 25/10, 40/25).
- Wing curvature and left–right asymmetry.

Expanded dashboard book (gate not retuned)

- RR and BF at 5 / 10 / 15 / 20 / 25 / 30 / 35 / 40 / 45 delta.
- Delta points: ATM plus each put and call node (POINT_25P, POINT_25C, …).
- Slopes: ATM → every wing, and every far-minus-near pair on each wing.
- Listed strikes STRIKE_K, with z taken from KBIN of k = ln(K/F).
- The k-bins themselves.

That expansion was built from the already-saved surfaces. Smiles were not refit and 3.0 was not retuned. The point was to stop missing a name the old 46-structure book never looked at. On 11 Sep 2026 the expanded book still has zero TRADE at the frozen gate.

Kind	Named count	What residual means
Risk reversal	9	Skew vs the ensemble skew
Butterfly	9	Smile convexity vs ensemble convexity
Delta point	19	That node’s IV vs ensemble IV
Slope	90	Wing-minus-belly (or ATM) vs ensemble
k-bin	14	Average residual in a moneyness bucket
Listed strike	varies by day	That K’s IV; history from its k-bin

Named catalog on the live desk is 141 structures plus listed strikes each day.

7. Historical z — how “unusual” is measured

A 3 basis-point residual can be huge for a quiet butterfly and noise for a wild wing. So each structure is standardised against **its own past**, using only dates ≤ t.

```
z_w(t)  =  [ residual(t)  –  mean(residual over the last w days, including t) ]
           /  std(residual over the same window)
Windows w ∈ {20, 60, 120, 252} trading days.
The official gate uses z_60.
```

Minimum periods are capped so a 20-day window does not demand 20+ observations before it exists. Percentile rank in the same window is stored as a second view. We also z-score the raw market value (the level), but the trade rule uses the **residual** z. A high ATM IV is a regime fact, not a cheap/expensive call.

For listed strikes the time series is the k-bin, not the integer K. Otherwise a 6600 strike would "look cheap" simply because spot has rallied and that K is now a deep put.

8. Ten definitions of cheap / expensive — none is a trade by itself

The research engine attaches many definitions so a reviewer can see *why* something looks odd. The dashboard still shows those votes as cards. The frozen decision is stricter than any single card.

#	Definition	Meaning
1	Structural	Sign of residual vs ensemble. Market below fair → cheap.
2	Historical residual z	Is this residual unusual vs this name's last 60 days?
3	Relative to surface	Neighbour / local residual, not just the point in isolation.
4	Relative to delta	Same structure historically (the z series).
5	Skew	RR or put/call skew residual.
6	Curvature	Butterfly / wing-curvature residual.
7	Regime	High/low ATM IV; RV vs IV. Event premium proxy only.
8	Event-adjusted	No calendar in the files; RV / IV−RV stands in.
9	Liquidity-adjusted	Residual shrunk when the vega/liquidity score is poor.
10	Executable	Residual after a modelled half-spread. No real bid/ask.

Agreement matters. Structural cheap + historically cheap is a candidate. Structural cheap + historically normal is "a small residual on a quiet name". Structural cheap + historically expensive is a contradiction → CANNOT DECIDE.

9. How the gate value 3.0 was decided

The gate g is the number of residual standard deviations required before we are willing to speak. Candidates tested **only inside walk-forward validation**: 1.0, 1.5, 2.0, 2.5, 3.0. The holdout was not allowed to vote.

Selection rule (exact)

- For each candidate g, keep validation rows with $|z_{60}| \geq g$ and at least 30 observations.
- Score = mean 5-day residual convergence on those rows. Convergence is −sign(residual) × future residual change, i.e. "did the residual shrink in the fade direction?"
- Pick the g with the highest mean convergence. That winner is **3.0**, mean convergence ≈ 0.0013 (about 0.13 vol points).
- Then freeze g. Refit the ridge / logistic models on all research dates. Never touch g after reading 11 Sep.

Is 3.0 "fair"?

Fair in the statistical sense: it was the walk-forward winner for *mean convergence given that you actually got a signal*. At $|z| \geq 3$ the research-period signalled book had about 96% residual-convergence hit rate. That sounds excellent until you look at two other facts.

- **Sample collapse.** Moving from 1.5 to 3.0 throws most days away. The original research book had only ~92 signalled rows (99 labelled) at 3.0. RR_25D alone had about 6 days. High hit rate on a tiny, selected sample is not the same as a robust strategy.
- **Economic size is small.** 0.13 vol points of mean fade is real statistically and tiny versus a 0.20 vol-point modelled IV half-spread plus price half-spreads that grow in the wings.
- **Cost-adjusted P&L is negative.** Mid-price fade can look positive; the conservative executable P&L does not. That is why NO TRADE is a first-class output.

- **1.5 remains an analysis slider** on the dashboard so you can inspect milder dislocations without pretending they cleared the frozen book.

A lower gate would be “fairer” if the goal were more conversation and more false positives. A higher gate would be “fairer” if the goal were almost never to speak. 3.0 is the research compromise: speak rarely, and only when the residual is historically extreme. It is not a proof that 3.0 will win on the next sample.

Gate	What you get	What you pay
1.0	Many flags, noisy	Mostly ordinary residuals; weak as a book
1.5 analysis	More names to inspect	Not the frozen production rule
2.0–2.5	Middle ground	Still a lot of small edges vs costs
3.0 frozen	Highest val mean-convergence, ~96% hit on the few signals	n collapses; exec P&L still negative

10. How a decision is made today

Dashboard rule (what you see on Vercel)

This is the rule the live desk uses. It does not retune anything. Changing the z-gate slider only re-applies the same logic.

```
Let R = residual,  z = z_60 of that residual.

CANNOT DECIDE  if any of:
    • z is missing (not enough history)
    • R is missing
    • the delta point was extrapolated more than 0.03
    • sign(R) and sign(z) conflict

NO TRADE       if the reading is clean but |z| < gate

TRADE          if |z| ≥ gate  and  sign(R) = sign(z)
                 and not extrapolated
    direction: CHEAP if R < 0, EXPENSIVE if R > 0
    intended action: fade (buy cheap / sell expensive residual)
```

Worked example — 25-delta risk reversal on 11 Sep 2026

Step	Number
Expiry / DTE / forward	2026-11-05 · 56 DTE · F ≈ 7699.96
IV 25-delta call / put	0.115918 / 0.167339
Market RR = IV_C – IV_P	–0.051421 (–5.14 vol points)
Ensemble fair RR	–0.051769
Residual R	+0.000348 (about +3.5 bp of vol)
z_60	+0.18
Signs	R > 0 and z > 0 — they agree, so not a conflict
|z| vs gate 3.0	0.18 ■ 3 → NO TRADE
Plain English	Slightly expensive vs the smile, historically ordinary

A forced ranking by raw residual would have invented a “best” name. The historical rule correctly refuses. On the expanded book the largest named |z| that day is about 1.38 (POINT_15P). Still NO TRADE.

Research-engine extras (stricter live-book filter)

The pipeline’s decide_trade can also refuse a name that cleared the z-gate if model dispersion > 3 vol points, liquidity score < 0.25, predicted convergence probability < 0.5, or predicted edge after costs ≤ 0. Those filters matter on days that *do* clear |z| ≥ 3. On 11 Sep nothing reached the gate, so they never came into play.

11. Predictive models — used as a check, not as the judge

Walk-forward models try to forecast 5-day residual convergence from features known at t. Primary model: Ridge (standardised features). Secondary: logistic regression for P(convergence > 0). Elastic net, random forest and gradient boosting exist in the library but the frozen pair is ridge + logreg.

Features (only those present, never future IV or future RV)

- residual, price residual, z_20 / z_60 / z_120 of residual and of market level
- model dispersion, confidence = |residual| / dispersion, liquidity score
- actual DTE, |DTE−60|
- ATM IV, RV_20, RV_60, IV−RV_20
- spot return 5d / 20d, 20-day drawdown

Out-of-sample information coefficient flipped sign across folds (–0.17 to +0.16). Sign-hit ranged 47%–75%. There is **not** a stable mapping from the feature list to residual change. That is an independent reason the desk is allowed to say NO TRADE even when a residual is non-zero.

12. What the backtest actually showed

Horizon is 5 trading days. “Convergence hit” means the residual moved in the fade direction. Executable P&L uses modelled half-spreads (about 0.50 price points near ATM, plus up to 1.50 extra in the wings, and 0.20 vol points of IV spread) because the files have no bid/ask.

Book	n with P&L	Conv. hit	Mean conv.	Exec. hit	Mean exec P&L
All structures	9,167	63%	+1.9 bp vol	31%	negative
Signalled |z|≥3	80	96%	+14 bp vol	53%	still negative
Risk reversals	1,465	65%	+2.1 bp vol	~0%	negative
Butterflies	1,465	62%	+1.7 bp vol	52%	least-bad, Sharpe ~0.08
Verticals signalled	12	100%	+12 bp vol	0%	negative

Read this table as: statistics can be right and the trade still not be worth doing. Butterflies were the least-bad package in-sample. That is not a robustness claim.

On the expanded universe, using the same frozen 3.0 and no retune, there are 501 TRADE rows across 102 research dates (230 slopes, 140 listed-strike prints, 69 points, 31 RR, 27 BF, 4 k-bins). So the old 46-name book *was* missing names. The holdout still has none.

13. How fair / correct is this? An honest scorecard

“Correct” has to be split. A model can be correct as a **description of residuals**, correct as a **forecast of fade**, and still be incorrect as a **money machine**.

Question	Verdict	Evidence
Is the surface construction honest?	Yes	PCP forward, abs(delta), OTM IV, logged cleaning, one expiry.
Is “fair IV” a unique truth?	No	Five models; we take the median. Dispersion is shown, not hidden.
Do residuals mean-revert?	Weakly yes	corr(R, ΔR_5d) = –0.44. Size is small.
Does |z|≥3 pick better fades?	Yes, in-sample	Hit ~96% vs ~63% unfiltered. n is small.
Is 3.0 the only fair gate?	No	It won one walk-forward score. 1.5 is for inspection.
Does the ML add a stable edge?	No	IC changes sign by fold.
Do trades survive costs?	Not shown	Modelled exec P&L negative. No real bid/ask.
Was 11 Sep handled correctly?	Yes, given the rule	Largest |z| ~1.2–1.4. Gate 3.0 → NO TRADE.

Question	Verdict	Evidence
Did we overfit the holdout?	Protected	Gap days unused; holdout not used to pick g.
Multiple testing?	A real risk	Many names × horizons × gates. Holdout is the main brake.

What you should say if someone asks “how confident are you?”

- Confident that 11 Sep 2026 was not an extreme residual day versus this history.
- Confident that “below SVI” is not a trade on this slice — almost every strike is close to the smiles.
- Moderately confident that large residuals fade a few basis points of vol over five days.
- Not confident that fading them makes money after spreads.
- Not confident the same gate or the same names will win on the next year of data.

If the presentation needs a single number for “how correct”: treat the **direction of extreme residuals** as the supported claim (~high 60s% unfiltered, ~90s% at |z|≥3 in-sample) and treat **P&L** as unsupported. That is a more honest slide than a Sharpe.

14. Complete catalogue of methods used

A. Data and surface

- Last-print daily snapshot; calendar DTE; nearest-expiry selection with tie-break logging.
- Put-call parity ATM-neighbourhood median forward; DF = 1 because rate ≡ 0.
- OTM IV construction; total variance $w = \sigma^2 T$; $k = \ln(K/F)$.
- Black–Scholes price and IV inversion (audit only; vendor IV is the market).
- Static no-arbitrage checks: call decreasing in K, put increasing, discrete butterfly convexity.
- Liquidity score from vega (and optional hybrid with a quality bucket). No volume/OI available.

B. Smile / fair-value methods

- Quadratic polynomial in total variance.
- Smoothing spline in log-moneyness.
- Gatheral SVI on total variance, previous-day warm start on research days.
- SSVI *slice* (θ, ρ, ϕ) — not a multi-expiry SSVI surface.
- Hagan SABR with β fixed at 1 (lognormal).
- Robust ensemble = median; disagreement = cross-model standard deviation.
- Vega (default), equal, liquidity, or hybrid fit weights.

C. Interpolation and structures

- Linear interpolation of IV versus |delta|, max 0.03 extrapolation.
- Risk reversals, butterflies, delta points, ATM, skews, slopes, verticals, wing curvature.
- Price-space residuals via BS prices on interpolated strikes where defined.
- Listed-strike time series via 0.02 log-moneyness bins.

D. Statistics and signals

- Causal rolling mean / std / z / percentile at 20, 60, 120, 252 days.
- Realized vol: close-to-close, windows 5–60, annualised $\sqrt{252}$; RiskMetrics EWMA $\lambda = 0.94$.
- Future labels at horizons 1, 3, 5, 10, 20 days (research dates only; label date ≤ train end).
- Ten cheap/expensive definitions listed in §8.
- Frozen z-gate search on walk-forward mean 5-day convergence, $n \geq 30$.

E. Predictive ML (research window)

- Walk-forward expanding window, 80-day minimum train, 20-day validation, 5-day embargo.
- Ridge regression ($\alpha = 1$) on 5-day convergence.
- Elastic net (available).
- Logistic regression for P(positive convergence).
- Random forest and gradient-boosting classifiers (available, not the frozen pair).
- StandardScaler inside the linear pipelines. Seed 7.
- Metrics: IC (Pearson of prediction vs realised), sign-hit, MAE.

F. Backtest and costs

- Fade the residual for 5 days on research dates only.
- Mid P&L vs conservative executable P&L.
- Cost model: 0.50 ATM price half-spread, +1.50 wing extra, 0.002 IV half-spread.
- Hit rate, mean/median P&L, Sharpe proxy, max-drawdown proxy — all labelled as proxies.

G. Validation design

- No random splits. No peeking at 11 Sep for thresholds.
- 9–10 Sep unused.
- Single frozen holdout. Subsequent days score that decision only.
- Multiple-testing warning recorded in the final report.

H. Software / delivery

- Python: pandas, numpy, scipy, scikit-learn, pytest.
- Modular pipeline under src/ (loader, clean, DTE, construction, models, structures, features, signals, backtest).
- Static dashboard (HTML/JS/CSS) + compact columnar panel.json.
- Private GitHub repo; Vercel static host.

15. How to present the dashboard in five minutes

- Open https://snp-60dte-iv-desk.vercel.app and click **Jump to 11 Sep 2026**. Gate defaults to 3.0 frozen.
- The count strip is the slide: TRADE / NO TRADE / CANNOT DECIDE for that day. On the holdout, TRADE is zero.
- Pick RR_25D. Show market vs ensemble (almost on top of each other) and $z \approx 0.18$.
- Switch the gate slider to 1.5 and explain that this is *analysis*, not a retune. Most names still do not clear it that day.
- Filter Kind = slope or listed_strike to show that the book is no longer just 46 names.
- Open a research date that does have TRADEs (102 such dates exist at gate 3.0) if you want to show what a cleared signal looks like — then remind the room those days were inside the sample that chose the gate.
- Close on the cost slide: 96% statistical hit, negative executable P&L, therefore NO TRADE is the designed output.

16. What this project refuses to claim

- That ATM IV of 13.45% on 11 Sep is cheap or expensive. That is a level.
- That any strike below SVI is a buy.
- That 25P/15P or any other vertical is “the best trade” after scanning many deltas.
- That walk-forward IC will stay positive.
- That modelled costs equal live SNP bid/ask.
- That SSVI here is a calendar-arbitrage-free surface.
- That the expanded universe was used to retune 3.0 — it was not.

17. Bottom line for the room

The model is a disciplined way to say whether a 60-DTE structure is cheap or expensive *relative to a multi-model smile and to its own history*. The frozen gate of 3.0 was chosen by walk-forward mean 5-day convergence, not by the holdout, and not by RMSE. That rule is statistically conservative and economically humble: it found residual fade and did not find a cost-surviving book. On 11 September 2026 the correct output of that frozen rule, even after opening every RR, butterfly, slope and strike, is **NO TRADE**.

Training 2025-06-05 → 2026-09-08. Gap 9–10 Sep unused. Holdout 11 Sep 2026 untouched. Do not retune the gate after reading this briefing or after clicking the dashboard.